AsmFEA verification report

Classical-mechanics checks of the full chain, SolidWorks to report, and of the CalculiX solver

07 October 2026 | AsmFEA 0.4.0 | CalculiX 2.22 | Full-chain runs from SolidWorks

1 Summary

Users should satisfy themselves of AsmFEA's accuracy for their own work, by running their own tests against hand calculations or other trusted results, before relying on it.

13 / 13
comparisons within their acceptance criteria

+0.62 %
largest difference from the reference (C3, tip deflection)

101 / 101
CalculiX test examples matching their reference results

Case	Quantity	Reference	AsmFEA	Difference	Result
C1	Tip deflection (mm)	23.555	23.647	+0.39 %	PASS
C1	Bending stress at mid-length (MPa)	92.216	92.217	+0.001 %	PASS
C1	Mass (SolidWorks mass properties) (kg)	28.819	28.819	+0.000 %	PASS
C2	Mass (theory) (kg)	50.24	50.24	0.000 %	PASS
C2	Mass (SolidWorks mass properties) (kg)	50.24	50.24	0.000 %	PASS
C2	Total reaction (= weight) (N)	492.85	492.85	0.000 %	PASS
C2	Mid-span deflection (mm)	0.14382	0.14376	-0.037 %	PASS
C2	Mid-span bending stress (MPa)	2.8878	2.8888	+0.035 %	PASS
C3	Tip deflection (mm)	23.555	23.701	+0.62 %	PASS
C3	Bending stress at z = 1500 (MPa)	46.108	46.108	+0.001 %	PASS
C3	Mass (SolidWorks mass properties) (kg)	28.819	28.819	+0.000 %	PASS
C4	Lowest buckling load (Euler) (N)	156,081	155,434	-0.41 %	PASS
C4	Mass (SolidWorks mass properties) (kg)	86.458	86.458	+0.000 %	PASS

6 of 6 method checks for joints and notches (M1-M4, on hand-written models) pass; see section M.

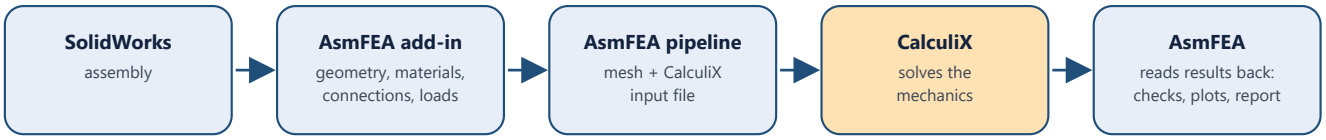
Difference = (AsmFEA − reference) / reference, sign kept; the bar shows how much of the acceptance criterion it uses. The reference is the classical solution, except the mass rows marked SolidWorks. Criteria: deflection, stress and buckling load 5 %, reactions 1 %, mass 0.5 %.

This report compares AsmFEA's results with classical hand calculations on a few simple cases, and checks the CalculiX solver against its own test examples and two standard NAFEMS benchmarks. It is a comparison between methods, not an approval of any particular analysis.

SOLIDWORKS Simulation results for the static cases can be added (SOLIDWORKS Premium, linear static). SOLIDWORKS Premium has no buckling study (linear buckling needs the separately licensed SOLIDWORKS Simulation Professional), so C4 is compared with theory alone: AsmFEA can run this analysis and SOLIDWORKS Premium cannot.

2 How the analysis works

> **The add-in does not calculate stress. It reads the geometry, materials, connections and loads from SolidWorks, builds a finite element model, and passes it to CalculiX. CalculiX solves it. AsmFEA then reads the results back and reports them. A model too large for CalculiX to hold in the computer's memory is solved instead by AsmFEA's own hybrid solver (section 3b), which reads the same input file and writes its results in the same form.**



The add-in prepares the model and reports the results; the stresses are calculated by CalculiX.

Two things are therefore shown separately: that the solver solves a correctly posed model accurately (CalculiX alone), and that AsmFEA poses each model correctly and reports it correctly (the full chain from SolidWorks).

3 The solver

CalculiX is a free, open-source finite element program by Guido Dhondt (solver) and Klaus Wittig (pre- and post-processor), distributed under the GNU GPL. AsmFEA runs it as a separate program. CalculiX has no formal certification and no vendor verification manual of the kind published for the large commercial codes; it is not described here as certified or validated by any body. What exists is its developer's own test examples and published benchmark comparisons, and both were run on the exact binary AsmFEA uses.

Binary	C:\Tools\PrePoMax v2.6.0\Solver\ccx_dynamic.exe (CalculiX Version 2.22, Copyright(C) 1998-2024 Guido Dhondt; You are using an executable made on Sun Aug 4 19:44:24 2024; 'ccx -v' prints 'This is Version 2.22')
MD5	e796e1851bc7c869474f6a9f9ece3a3b: the same file as in the author's calculix_2.22_4win.zip, which the AsmFEA installer downloads
Official test examples	**101 of 101** decks match their reference results (official compare scripts; a value fails beyond 0.1 % of the largest value in its output block)
With PARDISO, AsmFEA's solver	**100 of 101**; the exception tests the iterative Cholesky solver, whose reference values carry that solver's convergence error

NAFEMS LE1, elliptic membrane: σyy at point D, target 92.7 MPa

Element	Mesh	σyy at D (MPa)	vs target
CPS8	6 elements	81.63	-11.94 %
CPS8	24 elements	89.92	-3.00 %
CPS8	96 elements	93.01	+0.33 %
CPS8	384 elements	93.35	+0.70 %
CPS8	1,536 elements	93.18	+0.52 %
CPS8	6,144 elements	93.06	+0.38 %

NAFEMS LE10, thick plate: σyy at point D, target -5.38 MPa

Element	Mesh	σyy at D (MPa)	vs target
C3D10	2,680 elements, 13,207 equations	-5.498	+2.20 %
C3D10	19,328 elements, 85,812 equations	-5.385	+0.089 %
C3D10	127,732 elements, 540,539 equations	-5.378	-0.043 %

Targets: NAFEMS TNSB Rev. 3 (1990), quoted via bConverged (the NAFEMS document itself is not freely available). The official test set is thin on quadratic tetrahedra, wedges, gravity and buckling; the LE10 runs and the full-chain cases cover those.

Sources

Source	What it shows	Link
CalculiX home page	Authors, history, licence, releases	https://www.calculix.de
CalculiX 2.22 manual, section 11, verification examples	The test examples that ship with the solver	https://www.dhondt.de/ccx_2.22.pdf
Official examples repository	Official test examples, including the structural set	https://github.com/calculix/examples
NAFEMS LE1, elliptic membrane (bConverged)	CalculiX 1.5 against the NAFEMS target and ANSYS	http://www.bconverged.com/benchmarks/le1.php
NAFEMS LE10, thick plate (bConverged)	The same comparison for a 3-D solid benchmark	http://www.bconverged.com/benchmarks/le10.php
Parametric NAFEMS LE10 study (seamplex)	CalculiX among several solvers converging on LE10 (linked report)	https://seamplex.com/feenox/tests/nafems/le10/
Independent validation suite	Third-party decks with closed-form references (small, informal)	https://github.com/larsmei/calculix-validation
Theory reference	G. Dhondt, The Finite Element Method for Three-Dimensional Thermomechanical Applications, Wiley, 2004, ISBN 0-470-85752-8	-
NAFEMS on code verification	Why benchmark testing is the accepted way to verify an analysis code	https://www.nafems.org/community/the-analysis-agenda/code-verification/

SOLIDWORKS Simulation product page	Which analyses each SOLIDWORKS package includes	https://www.solidworks.com/product/solidworks-simulation

All links were checked on the date of this report.

3b Large models: the hybrid solver

CalculiX solves the whole model as one system of equations, and its memory grows with the model: about 8 GB per million unknowns. A model that would not fit is no longer refused. AsmFEA solves it with its own hybrid solver: each part is solved on its own and reduced to its joints (static condensation), the joints of all parts are solved together, and each part's full result is then recovered. Memory is set by the largest part, not the whole model, and parts are solved side by side within the memory the computer has free. Each joint between parts moves as a smooth field (polynomials of degree up to 4 over the joint area) rather than node by node, which is the one approximation it adds. A model CalculiX can hold is still solved by CalculiX, so every case in section 5 is a CalculiX result; the report of a study solved by the hybrid solver says so in its assumptions.

- Each part is assembled and factorised alone (Intel MKL PARDISO, the same direct solver CalculiX uses); a part with many joints is condensed with PARDISO's Schur complement.
- Parts are solved side by side within a memory budget (the free memory less a reserve); results that do not fit wait on disk.
- The solve checks itself: reactions must balance the applied load, and a model that is not held together is refused (as with CalculiX) rather than returning numbers.
- Decks it does not solve (buckling, contact, preloaded bolts, rigid couplings) stay with CalculiX.

Model	Result
Benchmark truss S: 72 welded SHS members, 4.0 M unknowns	Against CalculiX on the same input: largest displacement -0.64 %, worst node within 0.70 % of it, 95th percentile stress 0.00 %, reactions equal. One peak stress at a joint -12.7 %. Hybrid 159 s and 15 GB; CalculiX 193 s and 30 GB.
Welded tube chassis: 24 parts (rectangular tubes, plates and brackets), 90 bonded joints, 0.34 M unknowns	Against CalculiX: largest displacement -3.1 %, 95th percentile stress -3.6 %, reactions equal.
Benchmark truss L: 296 welded SHS members, 17.2 M unknowns (too large for CalculiX in 64 GB)	Tip deflection under 3 g self-weight 54.0 mm, against a hand-calculated lower bound of 52.4 mm (chord bending only; the lacing adds shear deflection) and SimSolid's 53 mm at 4 passes. Reactions equal the weight (8113 N). Hybrid 15 min at 30 GB; SimSolid 52 min.

Times measured on an 8-core Intel i9 workstation with 64 GB of memory. Peak stresses right at a joint differ more than stresses elsewhere, because the joint's smooth field spreads the load a little differently from CalculiX's node-by-node tie; the factor of safety in the report already excludes the band next to a tie (section 5b).

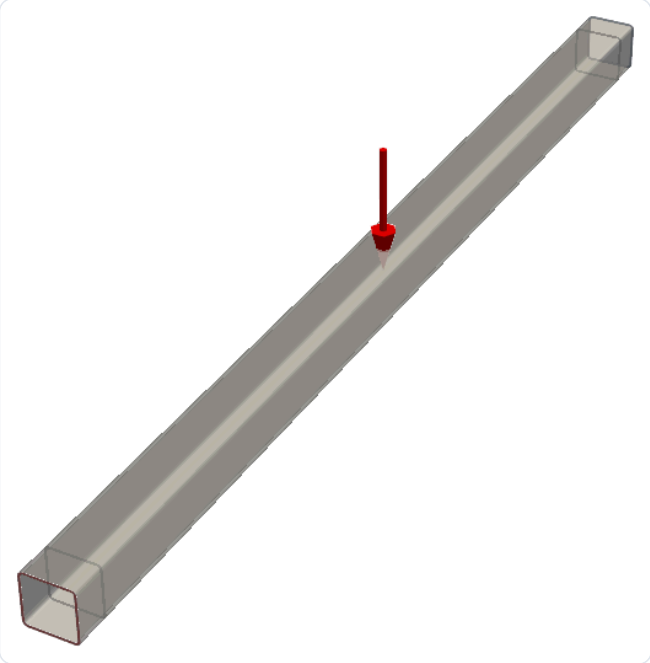
4 Method

- Full chain: every case starts as a SolidWorks model and runs through the add-in, the pipeline and the report generator, with nothing edited by hand in between. Loads and fixtures were set through the add-in on faces selected in SolidWorks.
- Theory first: each classical calculation was written and committed before the analysis for that case was run, so neither could be adjusted to agree with the other.
- Like with like: slender members (length at least 20 times depth); section properties from the modelled section with its corner radii; shear deflection included; stresses read at least one section depth clear of supports and load points; nominal stress compared with nominal stress.
- Default settings: every case at the default mesh fidelity (standard), which runs a base and a refined mesh and reports the refined one; C1 also at coarse and fine.
- Read-outs: each value is the mean of CalculiX's nodal results over a node set defined by coordinates, stated with each result; global values (mass, reaction, buckling factor) come from AsmFEA's results.
- No tuning: no mesh setting was changed per case to obtain a pass.
- SOLIDWORKS Simulation, where supplied, is a second opinion, not an acceptance criterion: two finite element codes agreeing is weaker evidence than agreement with theory, since they share the same kind of assumptions.

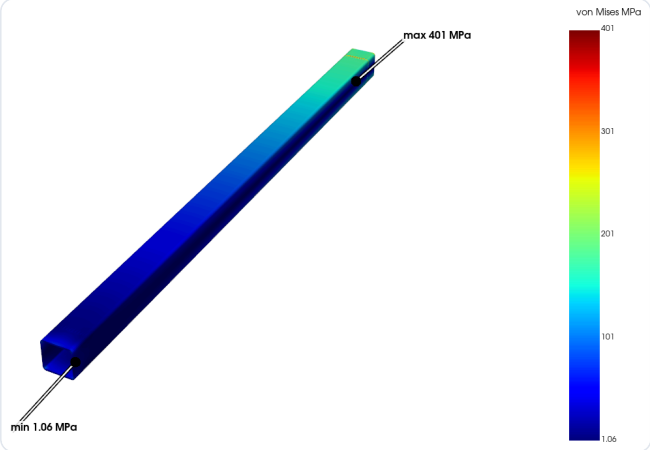
Quantity	Criterion
Deflection and rotation	within 5 % of theory
Nominal stress at the stated read-out point	within 5 %
Buckling load	within 5 %
Reactions	within 1 % of applied load
Mass	within 0.5 % of the SolidWorks mass properties

C1 Hollow-section cantilever with an end load

Exercises	Weldment member meshed as a swept member; fixed face; force load
Model	SHS 100 x 100 x 5 (outer corner radius 10 mm, inner 5 mm), 2000 mm long, as a SolidWorks weldment structural member. Material S355 verification (E 210 GPa, nu 0.3).
Restraints and loads	Root end face (z = 0) fixed: all translations of its nodes held. A 5 kN force in -Y spread over the tip end face (z = 2000).
Mesh	default fidelity (standard); reported mesh refined, 679,932 degrees of freedom
Convergence check	largest change between base and refined meshes 0.000 %



Model as analysed: fixtures blue, loaded faces red



Result: von Mises stress

Comparison

Quantity	Reference	AsmFEA	Difference	Result	Read-out
Tip deflection (mm)	23.555	23.647	+0.39 %	PASS	mean -Uy over the tip end face (z = 2000.0) (260 nodes)
Bending stress at mid-length (MPa)	92.216	92.217	+0.001 %	PASS	mean σzz over the top outer face, |x| ≤ 40, y = 50, z = 1000.00 (28 nodes)
Mass (SolidWorks mass properties) (kg)	28.819	28.819	+0.000 %	PASS	total mass of the mesh

The tip deflection is 0.39 % above beam theory (Euler-Bernoulli plus Timoshenko shear) at the default mesh and rises slightly with refinement, to 0.61 % on the fine mesh: the solid model represents what beam theory leaves out, chiefly the restraint and local flexibility at the clamped end face and the real distribution of shear in the thin walls (the theory uses an approximate shear area). Mid-length stress agrees to 0.001 % at every mesh. The convergence check is flagged because the peak stress sits at the clamped root, a stress singularity that grows with refinement; the read-out points are far from it.

▼ **Classical calculation (written before the run)**

```
================================================================================
C1  SHS 100x100x5 cantilever, end load
Reference: Roark 8th ed. Table 8.1 case 1a; Timoshenko shear deflection
================================================================================
  b = h          =              100 mm         outer width and depth
  t              =                5 mm         wall
  r_out          =               10 mm         outer corner radius, 2 t (EN 10219)
  L              =             2000 mm
  P              =             5000 N          end load, -Y
  E              =           210000 MPa
  nu             =              0.3 -

-- Section properties from the modelled section (OpenCascade, corner radii included)
  A              = area of the section face
                 =          1835.62 mm^2
  I              = second moment about x of the section face
                 =      2.71102e+06 mm^4
```

```
  A (sharp)      = b^2 - (b - 2t)^2  [cross-check, no radii]
                 =             1900 mm^2
  I (sharp)      = (b^4 - (b - 2t)^4) / 12  [cross-check, no radii]
                 =      2.86583e+06 mm^4
  G              = E / (2 (1 + nu))
                 =          80769.2 MPa
  A_s            = A h / (b + h)
                 =           917.81 mm^2

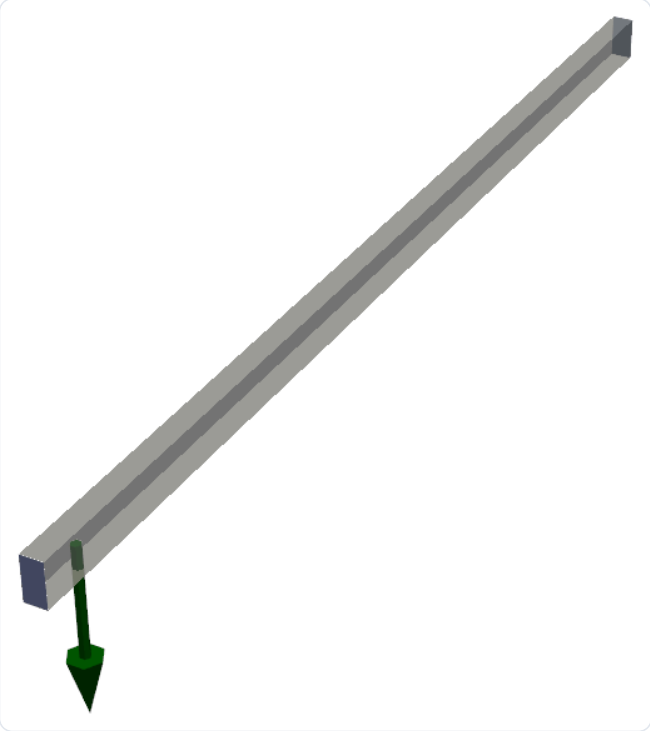
-- Tip deflection
  delta_bend     = P L^3 / (3 E I)
                 =            23.42 mm
  delta_shear    = P L / (G A_s)
                 =         0.134897 mm
  delta          = delta_bend + delta_shear
                 =          23.5549 mm

-- Bending stress at mid-length, outer fibre
  z              =             1000 mm         read-out position
  M              = P (L - z)
                 =             5e+06 N mm
  sigma          = M (h / 2) / I
                 =          92.2162 MPa
  sigma_root     = P L (h / 2) / I  [largest stress, at the fixed end; must stay below yield]
                 =          184.432 MPa

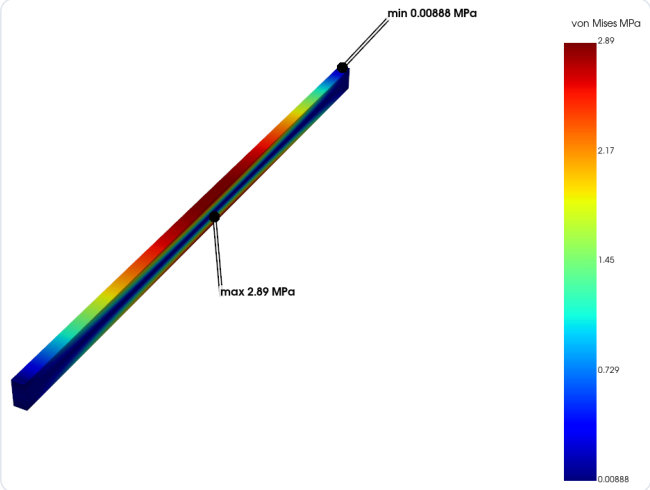
Results:
  tip_deflection_mm            23.5549
  midlength_bending_stress_MPa 92.2162
  A_mm2                        1835.62
  I_mm4                        2.71102e+06
```

C2 Solid rectangular bar, simply supported, under self-weight

Exercises	Quadratic tetrahedral mesh; gravity; mass; reactions
Model	Solid bar 40 mm wide (x) by 80 mm deep (y), 2000 mm long. Material S355 verification (rho 7850 kg/m3).
Restraints and loads	Both end faces pinned: each face is rigid and its centroid is held in x, y and z while the face rotates freely. Rotation about the bar axis (twist) is also held at z = 0, because two plain pins leave the bar free to spin about its axis (AsmFEA's mechanism check refused the first run without it; twist does not affect the bending compared). Self-weight, g = 9810 mm/s^2 in -Y.
Mesh	default fidelity (standard); reported mesh refined, 622,596 degrees of freedom
Convergence check	largest change between base and refined meshes +0.001 %



Model as analysed: fixtures blue, loaded faces red



Result: von Mises stress

Comparison

Quantity	Reference	AsmFEA	Difference	Result	Read-out
Mass (theory) (kg)	50.24	50.24	0.000 %	PASS	total mass of the mesh
Mass (SolidWorks mass properties) (kg)	50.24	50.24	0.000 %	PASS	total mass of the mesh
Total reaction (= weight) (N)	492.85	492.85	0.000 %	PASS	total vertical reaction over both supports (results.json support_reaction)
Mid-span deflection (mm)	0.14382	0.14376	-0.037 %	PASS	mean -Uy over the whole section at z = 1000.00 (9 nodes)
Mid-span bending stress (MPa)	2.8878	2.8888	+0.035 %	PASS	mean σzz over the bottom face (y = -40), z = 1000.00 (2 nodes)

Mass, reaction and bending agree to better than 0.05 %. The total reaction is AsmFEA's support reaction, which includes the self-weight lumped on the supported faces (found and corrected during this verification).

▼ **Classical calculation (written before the run)**

```
================================================================================
C2  Solid bar 40 x 80, simply supported, self-weight
Reference: Roark 8th ed. Table 8.1 case 2e; Timoshenko
================================================================================
  b             =              40 mm
  h             =              80 mm
  L             =            2000 mm
  E             =          210000 MPa
  nu            =             0.3 -
  rho           =        7.85e-09 t/mm^3     7850 kg/m^3
  g             =            9810 mm/s^2
  A             = b h
```

```
                =             3200 mm^2
  I             = b h^3 / 12
                =      1.70667e+06 mm^4
  G             = E / (2 (1 + nu))
                =          80769.2 MPa

-- Mass and reactions
  mass          = rho A L
                =          0.05024 t
  mass_kg       = mass x 1000
                =            50.24 kg
  w             = rho g A
                =         0.246427 N/mm
  R             = w L / 2  (each support)
                =          246.427 N

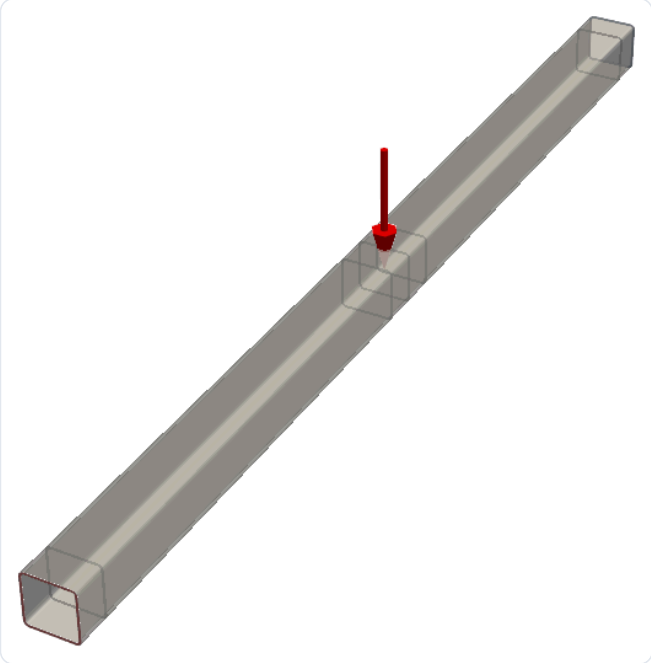
-- Mid-span deflection
  delta_bend    = 5 w L^4 / (384 E I)
                =         0.143245 mm
  delta_shear   = w L^2 / (8 (5/6) G A)
                =      0.000572063 mm
  delta         = delta_bend + delta_shear
                =         0.143817 mm

-- Mid-span bending stress, bottom fibre
  sigma         = w L^2 (h / 2) / (8 I)
                =          2.88782 MPa

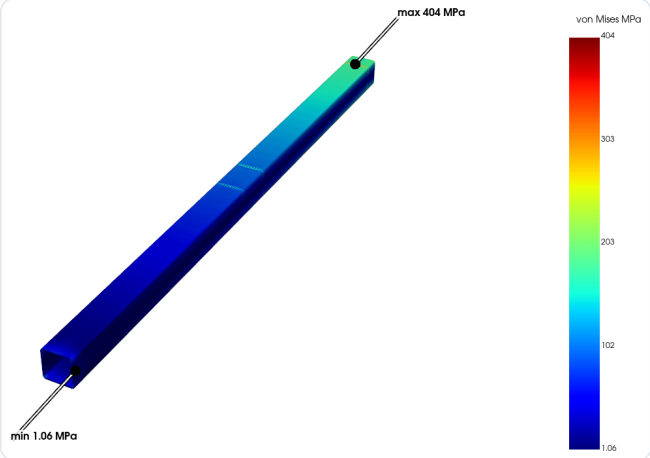
Results:
  mass_t                      0.05024
  reaction_each_N             246.427
  midspan_deflection_mm       0.143817
  midspan_stress_MPa          2.88782
```

C3 The C1 cantilever built from two members tied end to end

Exercises	Resolver: touching faces of two parts found and bonded (tie); swept member mesh
Model	The C1 section in two separate parts of 1000 mm each, touching at z = 1000, mated end to end (end faces coincident, two side faces flush).
Restraints and loads	Root end face of the first member (z = 0) fixed; 5 kN in -Y over the tip end face of the second (z = 2000). The resolver bonds the touching faces at z = 1000.
Mesh	default fidelity (standard); reported mesh refined, 691,416 degrees of freedom
Convergence check	largest change between base and refined meshes 0.000 %



Model as analysed: fixtures blue, loaded faces red



Result: von Mises stress

Comparison

Quantity	Reference	AsmFEA	Difference	Result	Read-out
Tip deflection (mm)	23.555	23.701	+0.62 %	PASS	mean -Uy over the tip end face (z = 2000.0) (260 nodes)
Bending stress at z = 1500 (MPa)	46.108	46.108	+0.001 %	PASS	mean σzz over the top outer face, |x| ≤ 40, y = 50, z = 1500.00 (28 nodes)
Mass (SolidWorks mass properties) (kg)	28.819	28.819	+0.000 %	PASS	total mass of the mesh

> The tie between the two members adds 0.23 % to the C1 tip deflection; stress at z = 1500 agrees to 0.001 %.

▼ **Classical calculation (written before the run)**

```
================================================================================
C3  C1 cantilever as two tied members
Reference: As C1: Roark 8th ed. Table 8.1 case 1a; Timoshenko
================================================================================
  b = h          =              100 mm
  t              =                5 mm
  r_out          =               10 mm
  L              =             2000 mm         two parts of 1000
  P              =             5000 N          end load, -Y
  E              =           210000 MPa
  nu             =              0.3 -
  A              = area of the section face
                 =          1835.62 mm^2
  I              = second moment about x
                 =      2.71102e+06 mm^4
  G              = E / (2 (1 + nu))
                 =          80769.2 MPa
  A_s            = A h / (b + h)
                 =           917.81 mm^2
  delta          = P L^3 / (3 E I) + P L / (G A_s)
                 =          23.5549 mm
  z              =             1500 mm         read-out position, clear of the tie
```

```
  sigma          = P (L - z) (h / 2) / I
                 =          46.1081 MPa

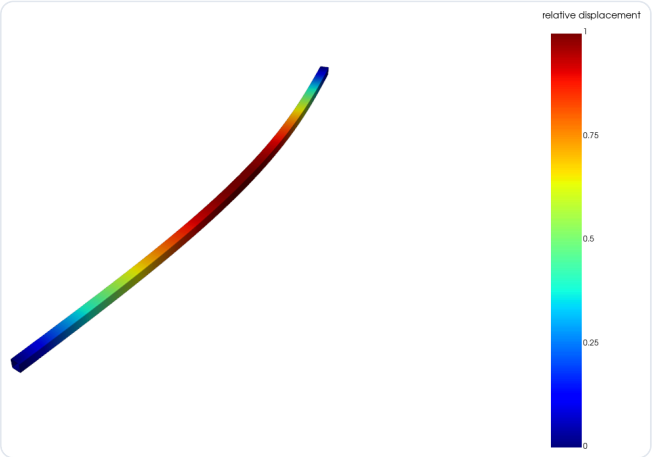
Results:
  tip_deflection_mm            23.5549
  bending_stress_z1500_MPa     46.1081
```

C4 Pin-ended hollow-section column

Exercises	Linear buckling; pinned ends with selected degrees of freedom
Model	The C1 section as a 6000 mm column along z.
Restraints and loads	Bottom end face pinned (rigid face, centroid held in x, y and z; twist about z held). Top end face pinned (rigid face, centroid held in x and y, free to move along z). 1 kN axial compression on the top face, the reference load the buckling factor scales.
Mesh	default fidelity (standard); reported mesh refined, 2,032,020 degrees of freedom
Convergence check	largest change between base and refined meshes +0.005 %



Model as analysed: fixtures blue, loaded faces red



Result: buckling mode 1

Comparison

Quantity	Reference	AsmFEA	Difference	Result	Read-out
Lowest buckling load (Euler) (N)	156,081	155,434	-0.41 %	PASS	lowest buckling factor x 1000 N (results.json)
Mass (SolidWorks mass properties) (kg)	86.458	86.458	+0.000 %	PASS	total mass of the mesh

> The buckling load is 0.4 % below Euler and 0.2 % below Euler with the shear correction (155.75 kN); the two lowest modes are equal, as for a square section.

▼ Classical calculation (written before the run)

```
================================================================================
C4  Pin-ended SHS 100x100x5 column, 6 m
Reference: Euler; Timoshenko & Gere, Theory of Elastic Stability, ch. 2
================================================================================
  b = h          =              100 mm
  t              =                5 mm
  r_out          =               10 mm
  L              =             6000 mm
  E              =           210000 MPa
  nu             =              0.3 -
  P_ref          =             1000 N          applied axial load
  A              = area of the section face
                 =          1835.62 mm^2
  I              = second moment (equal about both axes)
                 =      2.71102e+06 mm^4
  r_g            = sqrt(I / A)
                 =          38.4304 mm
  lambda         = L / r_g
                 =          156.126 -
  b / t          = b / t
                 =               20 -
  G              = E / (2 (1 + nu))
                 =          80769.2 MPa
  A_s            = A h / (b + h)
                 =           917.81 mm^2
  P_cr           = pi^2 E I / L^2
                 =           156081 N
  P_cr,shear     = P_cr / (1 + P_cr / (G A_s))
                 =           155753 N
  shear effect   = P_cr,shear / P_cr - 1
                 =      -0.00210105 -
  sigma_cr       = P_cr / A  [must stay below yield for elastic buckling]
                 =          85.0289 MPa

Results:
  P_cr_N                      156081
  P_cr_shear_corrected_N      155753
  buckling_factor             156.081
```

M Methods for joints and notches

A sharp edge bonded to another part, or a sharp notch, gives a stress that grows without limit as the mesh is refined: the peak there is a property of the model, not of the design. AsmFEA therefore classifies stresses before checking them against yield, using established methods. Each is listed with the situations it is there for and the evidence for it.

Method	What it does	Situations it is there for	Evidence
Bonding by proximity	Faces within the study's bond tolerance (default 0.5 mm), whether a gap or an overlap, count as touching and are bonded; parts need not be mated.	Parts placed rather than mated; small modelling gaps or overlaps between parts.	Automated tests: a 0.8 mm gap is left unbonded at the default and bonded at 1 mm; a 0.3 mm overlap is bonded without an interference warning.

Weldment joints	Inside one weldment, bodies meeting only along an edge are bonded along a 2 mm strip, and a member butted against another's side is bonded over its whole end, across the gap at the corner radii, as it is welded.	Rectangular and square tubes butted against another's side: only the flat between the radii touches, leaving a V gap a welder fills, which SimSolid and plain contact searches leave unbonded.	M3 below: the bonded joint deflects within 11 % of the same frame with its gaps weld-filled; without the rule it matches the open gaps.
Edge on a face	Where a plate's edge meets a face, the small side is tied onto the large one, and a tie uses only the nodes over the touching area.	Thin plates welded or bonded edge-on to tubes or plates; coarse meshes on the large face.	Found on the chassis (a plate floated); automated tests and the M1 check use this joint.
Structural stress at bonded joints	The joint is read from the forces it carries (Dong's mesh-insensitive structural stress, Battelle; the method in Verity): the forces at the bonded face of the member are turned back into a traction field and integrated along the joint into membrane, bending and shear: sigma = f / t + 6 m / t^2.	Plate edges and tube ends bonded to other parts, where the raw peak is singular.	M1 below: the average joint bending equals the hand answer exactly (statics), the worst point matches the plate's own stress at its root, and coarse and standard meshes agree to 1 %.
Membrane plus bending of member stresses	As in ASME VIII Div. 2 Part 5: at the most stressed points the stress is linearised along a line through the material; membrane plus bending is checked against yield, the peak above it is a notch effect for fatigue. In thin parts the line runs across the ligament from a notch's edge.	Holes, notches and internal corners in plates and tubes; local peaks above yield that a ductile material redistributes.	M2 below: the ligament membrane stress beside a hole is within 6 % of statics; membrane plus bending lies between it and the notch peak. On the chassis it was stable to 1-8 % between meshes where peaks moved up to 60 %.
Joint band and singular-peak flag	Member peaks are read at least 1.5 x the wall thickness (and 3 mm) from a bond's edge; with the convergence check on, a peak that keeps rising on the finer mesh while the deflection settles is flagged as a singular point and does not fail the check.	Remaining sharp corners and bond edges not covered above.	Automated tests (the flag's broken case); on the chassis the joint peak rose 52 % between meshes while the assessed value rose 14 %.
Wall thickness from the mesh	Where the model gives none, a part's wall thickness is measured from its mesh (rays through it from its largest faces).	Weldment profiles the add-in cannot read.	On the chassis: 1.5 mm measured on all 16 tubes (RHS 25 x 25 x 1.5).

Method checks

Hand-written models run through the pipeline (no SolidWorks), each against a hand answer. M4 checks the receiving-wall method, which is not used in normal runs (see Limitations).

M1 Structural stress at a bonded joint (Dong / Battelle)

Quantity	Reference	AsmFEA	Difference	Result	Read-out
Joint bending stress averaged over the joint (statics: 6 P L / b t^2) (MPa)	180	180	0.000 %	PASS	the recovered traction on the plate's bonded edge

Worst joint structural stress, coarse mesh against standard (MPa)	207.15	204.93	-1.07 %	PASS	the plate's bonded edge; peak of the moving window

M2 Membrane plus bending at a notch (ASME VIII-2 Part 5)

Quantity	Reference	AsmFEA	Difference	Result	Read-out
Ligament membrane stress (statics: N / 2 / ligament area) (MPa)	25	23.56	-5.76 %	PASS	stress classification line from the hole's edge across the ligament
Membrane + bending, below the notch peak (reference: the peak) (MPa)	55.231	30.236	-45.25 %	PASS	the same line

M3 Weldment butt joint bonded across corner radii

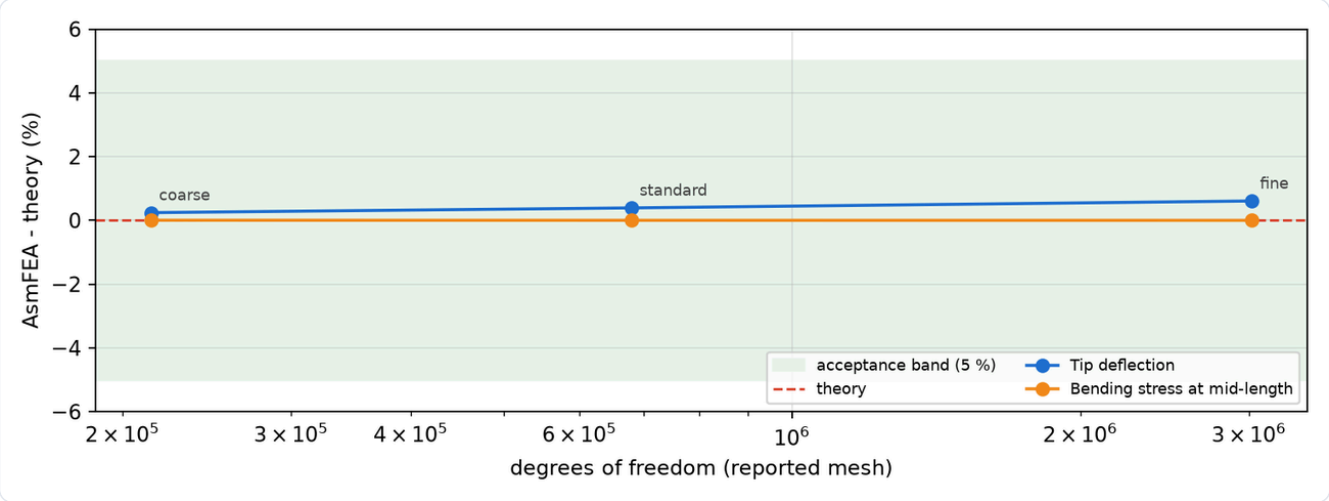
Quantity	Reference	AsmFEA	Difference	Result	Read-out
Post-top deflection: weldment bond against the frame fused with its V gaps weld-filled (mm)	0.97255	0.86913	-10.63 %	PASS	largest displacement (the post's top)

M4 Receiving wall of a bonded joint (research, off in normal runs)

Quantity	Reference	AsmFEA	Difference	Result	Read-out
Receiving-wall structural stress at the toe (clamped beam: 6 M / b t^2, M = N L / 8 - N t / 4) (MPa)	57	55.662	-2.35 %	PASS	the wall under the plate's bonded edge (research: off in normal runs)

6 Mesh sensitivity (C1)

Fidelity	Degrees of freedom	Tip deflection (mm)	vs theory	σ mid-length (MPa)	vs theory	Check: largest change	Flagged
coarse	214,116	23.612	+0.24 %	92.219	+0.003 %	+8.62 %	no
standard	679,932	23.647	+0.39 %	92.217	+0.001 %	0.000 %	no
fine	3,021,072	23.698	+0.61 %	92.215	-0.001 %	0.000 %	no



Difference from theory against degrees of freedom; green band: the 5 % criterion

The stress at the read-out point is converged to within a few thousandths of a percent at every mesh. The tip deflection rises slightly with refinement and is still settling, a little above beam theory (see the C1 comment), well inside the 5 % criterion. The tool's own convergence check compares each run's base and refined meshes and reports the largest change in peak stress or displacement: it flags the standard and fine runs, because the peak stress at the clamped root is a singularity that does not converge, and does not flag the coarse run, where that peak changed by 8.6 %, under its 10 % limit. This shows the check flags an unconverged peak; it also shows that a peak at a sharp clamp should be read with care whatever the mesh.

7 Limitations

- Linear elastic material, small displacements, linear buckling (no imperfections, no plasticity).
- Not covered by a case: separating contact and friction, preloaded bolts, revolute and sliding joints, weld beads drawn in SolidWorks, pressure loads, remote torque. The joint and notch methods are checked on hand-written models (M1-M4), not through SolidWorks.
- The wall of the part a member is bonded onto (the receiving side of a joint) is not assessed by the joint method: the method built for it passes its hand check (M4) but is not yet stable between meshes on tube joints with small footprints, so it is off in normal runs. Stresses there are read by membrane plus bending outside the joint band.
- Weld throats are not assessed; the joint method gives the stress in the members at the joint.
- Fatigue and dynamics are not part of AsmFEA's first release and are not verified.
- The cases in section 5 are solved by CalculiX. The hybrid solver, used only when CalculiX would not fit in memory, is checked by the comparisons in section 3b, not by its own full-chain cases.
- Four cases cannot cover every geometry; they test the parts of the chain listed under each case.
- Neither CalculiX nor AsmFEA is certified or validated by any body. This report compares two methods on a few cases; it is not an approval of any particular analysis. Users should satisfy themselves of AsmFEA's accuracy for their own work, by running their own tests against hand calculations or other trusted results, before relying on it.

8 Findings during verification

- C2's first run was refused by AsmFEA's mechanism check: two plain pinned ends leave a bar free to spin about its own axis. The model was corrected (twist held at one end); the classical calculation did not change.
- C2 showed that AsmFEA reported the reaction from CalculiX's RF output, which leaves out load applied directly to held nodes (here 0.10 %, the self-weight on the supported faces). AsmFEA now also reports the support reaction, the full load the supports carry, and its report's resultant forces use it. The analysis itself was not affected: its balance check already included that load.
- The automation calls used to set up and capture these models arrived on a background thread; the add-in now hands pane work to its own thread.
- On a real assembly (a welded tube chassis) the raw peak stress at bonded junctions grew with each mesh refinement (726 to 1105 MPa), a singularity. This led to the joint and notch methods below; the checks M1-M4 were written for them.
- On the same assembly, sheet parts with small cut-outs were meshed with fine tetrahedra (8 million unknowns) because the sweep mistook a narrow strip beside a cut-out for a thinner wall; and a plate meeting tube edges floated because the wrong side of each tie was chosen. Both were fixed (side walls left out of the thickness test; the small side of an edge-on-face tie is tied; ties claim only the nodes over the touching area).
- A weldment profile the add-in could not read left tubes without a wall thickness; AsmFEA now measures it from the mesh.

9 Appendix

Software versions

AsmFEA	0.4.0
CalculiX	2.22 (ccx_dynamic.exe, PARDISO via Intel MKL 2022.2.1), MD5 e796e1851bc7c869474f6a9f9ece3a3b
Gmsh	4.15.2
SolidWorks	SOLIDWORKS 2026 (34.2.1)
Python	3.13.5

▼ Inputs and generated CalculiX files (SHA-256)

Case	File	Bytes	SHA-256 (first 16)
C1	C1_shs_cantilever\sw\C1_Cantilever_FEA\Verification\model.json	1,631	`0f530271a03a8829`
C1	C1_shs_cantilever\sw\C1_Cantilever_FEA\Verification\study.json	3,457	`209e03890f7f7440`
C1	C1_shs_cantilever\sw\C1_Cantilever_FEA\Verification\parts\SHS_2000_Default_As_Machined_.step	45,739	`2f3661d33cd6fda4`
C1	C1_shs_cantilever\sw\C1_Cantilever_FEA\Verification\run\20261006-175913-2d2d\refined\study.inp	17,428,365	`45ebbfb3b099e2f3`
C2	C2_bar_self_weight\sw\C2_Bar_FEA\Verification\model.json	1,156	`251daa297902457e`
C2	C2_bar_self_weight\sw\C2_Bar_FEA\Verification\study.json	3,062	`82eb6c1d1a37e34c`
C2	C2_bar_self_weight\sw\C2_Bar_FEA\Verification\parts\Bar_40x80_Default.step	15,891	`cef1e8f0081e0396`
C2	C2_bar_self_weight\sw\C2_Bar_FEA\Verification\run\20261006-180033-48cf\refined\study.inp	19,302,217	`6f6266e28cd1bd2d`
C3	C3_tied_cantilever\sw\C3_Tied_FEA\Verification\model.json	11,176	`8264e4a31240e5d1`

C3	C3_tied_cantilever\sw\C3_Tied_FEA\Verification\study.json	3,425	`257955a53eeeb4fe`
C3	C3_tied_cantilever\sw\C3_Tied_FEA\Verification\parts\SHS_1000_Default_As_Machined_.step	45,738	`17a4ee9482332d94`
C3	C3_tied_cantilever\sw\C3_Tied_FEA\Verification\run\20261006-180134-aa02\refined\study.inp	17,482,878	`29264c18a3055801`
C4	C4_shs_column\sw\C4_Column_FEA\Verification\model.json	1,624	`744fb26d1ab6706e`
C4	C4_shs_column\sw\C4_Column_FEA\Verification\study.json	3,739	`9c3b37c6faffa817`
C4	C4_shs_column\sw\C4_Column_FEA\Verification\parts\SHS_6000_Default_As_Machined_.step	45,733	`f311ca1213dd494a`
C4	C4_shs_column\sw\C4_Column_FEA\Verification\run\20261006-180243-da8d\refined\study.inp	53,934,507	`0342258dbd870488`